GLENCORE

Data Standard

Maintenance Plan and Maintenance Item

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The suite of Master Data Standards provide clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Maintenance Plan and Maintenance Items Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP Maintenance Plans and Items at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5

- Identify the minimum requirements for creating, maintaining, and retiring Maintenance Plans and Items
- Provide the guiding principles to manage Maintenance Plans and Items Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master data maintenance teams in order to understand how the object needs to be maintained.
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to; Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions.
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard.
- Data migration team for the build of transformation and load tools.
- Transition to support teams to assist in the developing of user guides.

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

2. Data Object Description

2.1 Definition

A Maintenance plan:

- Is a mechanism for scheduling regular maintenance for physical assets (Equipment and Functional Locations)
- Allows for scheduling to be either:
    - counter based derived from time (e.g. days, weeks, months, years) or usage (e.g. hours, tonnes, cycles) triggered by measurement readings from a Measuring Point that is a counter (e.g. an hour meter).
    - or non counter based derived on condition or reading (e.g. pressure, flow, temperature etc) triggered by a non counter device (e.g. an instrument, meter etc) in conjunction with the system that defines the allowable range.
- Is an object that groups maintenance items.
- Sets the start date of the schedule of the maintenance tasks.
- Specifies factory calendar, if applicable.
- Specifies call horizon and scheduling parameters, such as scheduling period, sort field and shift factor.
- Triggers work orders creation in combination with task lists and maintenance items information.

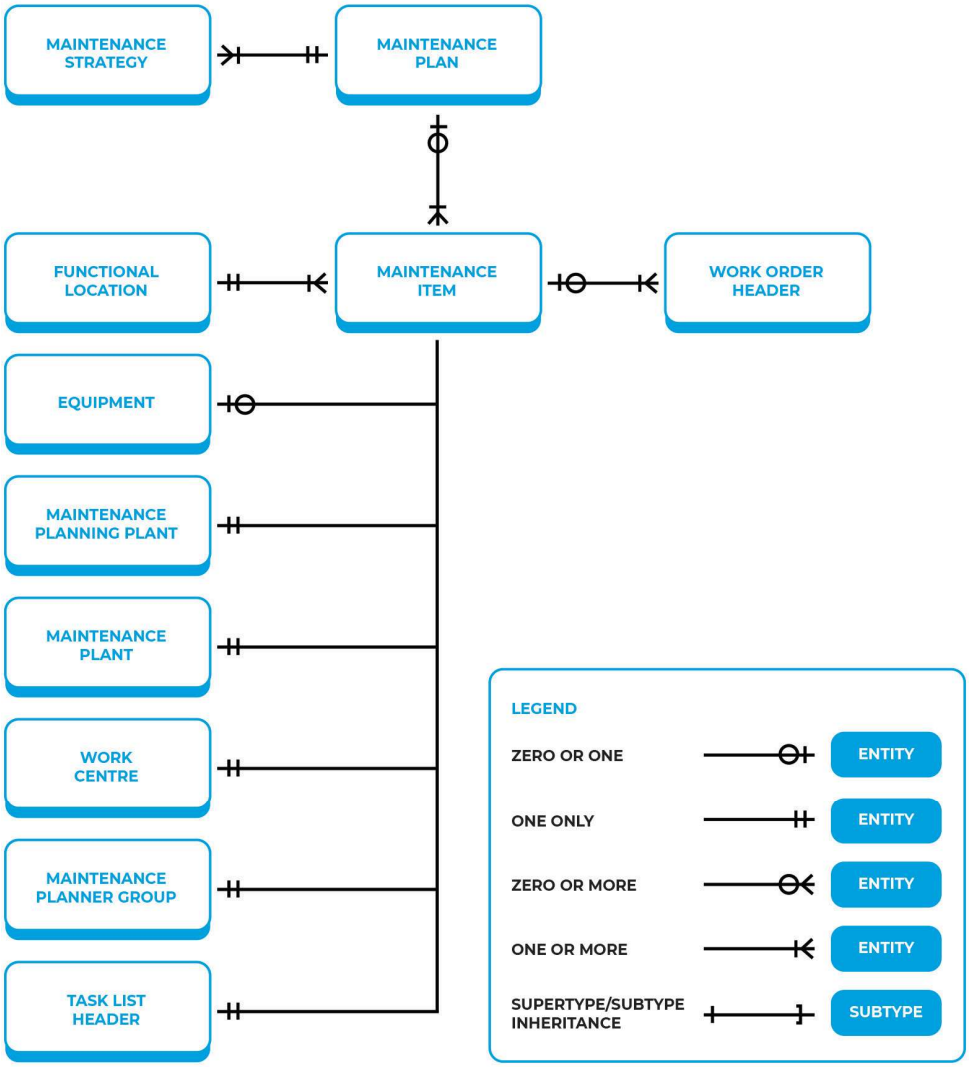
A Maintenance Item:

- Is a link to a physical asset on which work is to be performed and a Task List, which describes the work to be performed, the resources required, the work duration, and the frequency of execution.
- Provides the description of the task as short text in the work order header and priority.
- Describes which preventive maintenance tasks should take place regularly at a functional location or equipment.
- Provides the functionality to enable circuit maintenance tasks using the object list.

Maintenance orders are generated as a result of the scheduling of Maintenance Plans.

For strategy-based Maintenance Plans, the Maintenance Plan, the Maintenance Item(s), and the linked Task List(s) must all use the same Maintenance Strategy. A Maintenance Strategy in SAP is a grouping of frequencies (e.g. Weekly, 3 Monthly, Yearly). These frequencies are known as 'packages'.

2.2 OBJECT Relationship Diagram



3. Business Requirements

3.1 Business Use

The purpose of a Maintenance Plan is to transform the Maintenance Strategy into pre-defined, regularly occurring maintenance activities. Together with Task Lists, Maintenance Plans are an output of the Equipment Maintenance Strategy.

The Maintenance Plan collects the activities defined in a Task List, together with the physical asset on which the activity is to be performed and generates work orders according to the intervals defined in the Task List.

Maintenance Plans and Task Lists are also a key component of a zero-based budgeting process. The frequency defined in the Maintenance Plan allows the cost of an individual occurrence to be extended over a full year or more.

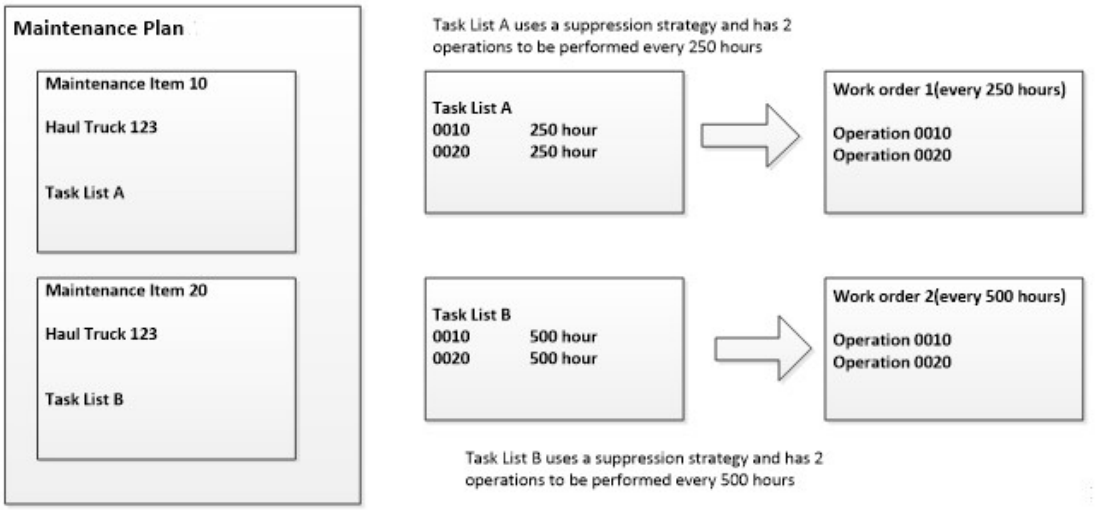
Process Area	Integration Points	Relationship
Plant Maintenance	▪ Asset Master Data Maintenance ▪ Equipment Maintenance Strategy Deployment ▪ Maintenance Budgeting ▪ Work Order Planning ▪ Shutdown Planning	▪ C,R,U,D ▪ C ▪ R ▪ R ▪ R
Finance	▪ Budget Development	▪ R

Relationship "C" – Create, "R" – Read, "U" – Update, "D" – Delete

3.2 Business design Rules & Criteria

The following business and design rules are to be applied to Maintenance Plans and Maintenance Items:

- Only time and usage-based Strategy and Multiple-Counter plans will be used.
- All regularly scheduled maintenance activities should be reflected in a Maintenance Plan.
- Structuring of Maintenance Plans should allow for the simplest interpretation of the types and frequencies of the activities to be performed. The example shown below allows the frequency of the task to be described in both the Maintenance Item text and the Task List header text. This requires discipline to ensure that the frequencies in the text match the assigned Maintenance Packages in the Task List operations, but enables at-a-glance information about the tasks in SAP list display transactions.



- All Maintenance Plan and Item descriptions will be in the relevant regional language. This supports standardising and sharing of Master Data across the organisation and is the endorsed position for the language of global IT systems.
- Maintenance Items will only be created for Equipment in the technical object structure. However, a Maintenance Item may be created for a level 5 Functional Location for circuit maintenance.

3.3 Restructuring Guidelines

When Equipment is no longer in productive use, either temporarily or permanently, any associated Maintenance Plans must be updated accordingly.

Scenario	Temporary decommissioning	Permanent decommissioning
All Equipment in a Maintenance Plan will be removed from productive use.	Set Maintenance Plan as inactive.	Set the deletion flag in the Maintenance Plan.
Equipment is part of a Maintenance Plan with other, still operational, Equipment. i.e. Maintenance Plan will still be active.	Remove Maintenance Item from the Maintenance Plan.	Remove Maintenance Item from the Maintenance Plan and attach it to the 'dummy' Maintenance Plan for archiving.

4. Detail Standard

4.1 Data Object Coding Standard

Both the Maintenance Plan and Maintenance Item will be internally (SAP system) generated numbers.

4.2 Attributes

4.2.1 Maintenance Plan Description

Application Definition	Description of the activities to be performed.
Application Guiding Principles	Free text describing the contents of the Maintenance Plan.
Guiding Principles	The description should follow a pre-defined sequence that clearly shows the purpose of the Maintenance Plan. All Maintenance Plan and Item descriptions will be in English.
Application Coding Convention	40 character alphanumeric
Naming Convention	Refer to Data Standard - General Configuration

4.2.2 Maintenance Plan Strategy

Application Definition	A rule for the sequence of planned maintenance and inspection tasks.
Application Guiding Principles	A Maintenance Strategy contains the maintenance packages that define the cycle in which the tasks must be performed, for example, every two months, every 5,000 km, every 500 operating hours. It contains additional scheduling parameters such as shift factor, preliminary and follow-up buffers, and hierarchy. Maintenance Strategies are obligatory in strategy plans.
Guiding Principles	Select the Maintenance Strategy appropriate to all the frequencies required for the Maintenance Plan.
Application Coding Convention	6 character Alphanumeric
Naming Convention	Refer to Data Standard - Maintenance Strategy

4.2.3 Maintenance Plan Sort Field

Application Definition	You can use the sort field for selections in the list editing of Maintenance Plans, scheduling overviews in list format, and deadline monitoring.
Application Guiding Principles	Not applicable
Guiding Principles	Used to link a Maintenance Plan to a site and to define whether it will be automatically scheduled or not.
Application Coding Convention	20 character alphanumeric
Naming Convention	****-** is a combination of the plant (4 characters) and department (2 characters) ******-**-ELE** — ****** ELECTRICAL PLANS** ****-**-ELE-INPLAN — **** ELECTRICAL INPLANNING PLANS** ****-**-MEC — **** MECHANICAL PLANS** ****-**-MEC-INPLAN — **** MECHANICAL INPLANNING PLANS** ****-**-SER — **** SERVICE PLANS** ****-**-SER-INPLAN — **** SERVICE INPLANNING PLANS** ****-**-SHT-MANUAL — **** MANUAL SHUTDOWN**

4.2.4 Maintenance Item Description

Application Definition	Description of the activities to be performed.
Application Guiding Principles	Free text describing the contents of the Maintenance Item.
Guiding Principles	The description should follow a pre-defined sequence that clearly shows the purpose of the Maintenance Item. As the Maintenance Item description is copied to the work order description, the description shall be consistent with the naming conventions for work order short text. All Maintenance Plan and Item descriptions will be in the relevant regional language.

Application Coding Convention	40 character alphanumeric
Naming Convention	Refer to Data Standard - General Configuration

4.2.5 Maintenance Item - Planning Plant

Application Definition	A plant in which maintenance tasks are planned and prepared. Maintenance Planning Plants are standard Plants that are indicated as Maintenance Planning Plants in SAP Configuration.
Application Guiding Principles	Not applicable
Guiding Principles	The Maintenance Planning Plant will default from the Equipment or Functional Location referenced in the Maintenance Item. The default should not be changed.
Application Coding Convention	4 character alphanumeric
Naming Convention	Refer to the Data Standard - General Configuration

4.2.6 Maintenance Item - Planner Group

Application Definition	A group of employees responsible for planning and processing maintenance tasks in a Plant.
Application Guiding Principles	Planner Groups or Maintenance Planning Groups must be defined separately for each Maintenance Planning Plant. Every piece of equipment and functional location can be allocated to a Maintenance Planning Group. Maintenance Planning Groups can also be used for evaluation purposes.
Guiding Principles	Each Planner Group represents a logical grouping for the task of planning and scheduling work orders. A Planner Group may represent one or more individuals whose primary function is to plan and schedule maintenance work orders, however a planner should normally only be responsible for one planner group. The value for Planner Group will default from the Equipment or Functional Location used in the Maintenance Item. It should be changed as required. For example, a Maintenance Plan specifically for a Shutdown would not normally have a Shutdown Planner Group defaulting from the Equipment or Functional Location.
Application Coding Convention	3 character alphanumeric

Naming Convention	Refer to Data Standard - General Configuration

4.2.7 Maintenance Item – Main Work Centre

Application Definition	Used at a higher level and represent the group responsible for work management activities. They are assigned to technical objects in the Functional Location Structure as well as the header of Work Orders, Task Lists and Maintenance Items.
Application Guiding Principles	Not applicable
Guiding Principles	The main Work Centre for a Maintenance Item represents the work group with the primary responsibility for the maintenance activity. The value will default from the Equipment or Functional Location used in the Maintenance Item. It should be changed according to the specific maintenance activities in the Maintenance Plan. For example, Operations may perform scheduled activities that are defined in a Maintenance Plan. In this case, the Work Centre should be changed to an Operations Main Work Centre.
Application Coding Convention	6 character alphanumeric
Naming Convention	Refer to Data Standard - General Configuration

5. Data Governance

5.1 Complete Field List

Legend for Attribute Table

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, CUKY=Currency key, UNIT=Measurement type, NUMC=Number)
- **Length –** Character length of field
- **MOCIND –** Data requirement (M=Mandatory, O=Optional, C=Conditional, I=Internally Populated, N=Not Used
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

5.1.1 Maintenance Plan Header

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPLA	WARPL	Maintenance Plan	Number that uniquely identifies a Maintenance Plan.	CHAR	12	M	Internally generated
MPLA	ERSDT	Date of creation	Date on which Maintenance Plan was created.	DATS	8	I	Internally populated field
MPLA	ERNAM	Name of Person who Created the Object	Userid of the person who created the Maintenance Plan.	CHAR	12	I	Internally populated field
MPLA	AEDAT	Changed On	Blank if Maintenance Plan has never been changed or the date on which the Maintenance Plan was last changed.	DATS	8	I	Internally populated field
MPLA	AENAM	Name of Person Who Changed Object	Blank if Maintenance Plan has never been changed or the userid of the person to last change the Maintenance Plan.	CHAR	12	I	Internally populated field
MPLA	WPTXT	Maintenance Plan Text	Short description of the contents of the Maintenance Plan.	CHAR	40	M	Refer to Data Standard - Short Text
MPLA	STRAT	Maintenance Strategy	Describes the unit and set of available frequencies that can be applicable to the Maintenance Plan. It also sets default scheduling parameters.	CHAR	6	M	Table Selection.
MPLA	ABRHO	Scheduling Period	Scheduling period is used to determine the length of time for which the system creates maintenance calls during Maintenance Plan scheduling.	NUMC	3	M	Set to 365. (recommended) The Maintenance Plan needs shall be ultimately considered when setting this value

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPLA	VSPOS	Shift Factor for Late Completion	The shift factor in the event of late completion of a maintenance task defines the percentage of the shift to be applied to the calculation of the next date when the variance is outside the tolerance range.	NUMC	3	O	A percentage value.
MPLA	VSNEG	Shift Factor for Early Completion	The shift factor in the event of early completion of a maintenance task defines the percentage of the shift to be applied to the calculation of the next date when the variance is outside the tolerance range.	NUMC	3	O	A percentage value.
MPLA	TOPOS	Tolerance for Late Completion (%)	Percentage tolerance for when the shift factor for late completion is to be applied.	NUMC	3	O	A percentage value.
MPLA	TONEG	Tolerance for Early Completion (%)	Percentage tolerance for when the shift factor for early completion is to be applied.	NUMC	3	O	A percentage value.
MPLA	SFAKT	Cycle modification factor	Increases or decreases the cycle length. A value greater than 1 increases the cycle length and a value less than 1 decreases it.	DEC	3	M	Default to 1
MPLA	STICH	Scheduling indicator	1 Time 2 Time - key date 3 Time - factory calendar	CHAR	1	M	As described
MPLA	HORIZ	Call horizon for Maintenance Plan calls	The call horizon determines when a maintenance order should be generated for a maintenance call.	NUMC	3	M	Any value in Days or %

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPLA	ANDOR	Indicator for OR or AND link	Used in multiple-counter plans to indicate whether one or both of the counters must be reached to generate a call. OR = earliest counter, AND = latest counter.	CHAR	1	C	
MPLA	BEGRU	Technical object authorization group	Allows update of Maintenance Plans to be restricted by	CHAR	4	N	Access of Maintenance Plans will not be restricted by the authorisation group. This is in line with the Security Principles.
MPLA	MPTYP	Maintenance Plan category	Indicator that determines which maintenance call object the system generates for a due maintenance call.	CHAR	2	M	Always use 'PM'. This is the default value and will generate a maintenance order.
MPLA	HUNIT	Unit in scheduling interval	Unit for the scheduling period	UNIT	3	M	See 'Scheduling Period'
MPLA	STADT	Start date	Date from which the system should start scheduling the Maintenance Plan.	DATS	8	C	Mandatory for time-based Maintenance Plans.
MPLA	SZAEH	Start counter reading	Counter reading from which the system should start scheduling the Maintenance Plan.	DATS	22	C	Mandatory for usage-based Maintenance Plans.
MPLA	CALL_CONFIRM	Completion Requirement	If you set this indicator, the system only generates the next call object once the preceding call object has been confirmed.	CHAR	1	O	
MPLA	PLAN_SORT	Sort field for Maintenance Plans	The sort field for selections in the list editing of Maintenance Plans, scheduling overviews in list format, and deadline monitoring.	CHAR	20	M	Table Selection. See Maintenance Plan Sort Field in the attributes section of this document.

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPLA	FABKL	Factory calendar key	Used to schedule a Maintenance Plan according to work days defined in the factory calendar.	CHAR	2	C	Table selection.

5.1.2 Maintenance Plan item

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPOS	WAPOS	Maintenance Item	A Maintenance Item describes which preventive maintenance tasks should take place regularly at a technical object or a group of technical objects.	CHAR	16	M	Internally generated number
MPOS	WARPL	Maintenance Plan	Number that uniquely identifies a Maintenance Plan.	CHAR	12	I	A Maintenance Item will have 0 or 1 Maintenance Plans associated with it. This field is automatically populated when a Maintenance Item is assigned to a Maintenance Plan.
MPOS	WSTRA	Maintenance Strategy	Describes the unit and set of available frequencies that can be applicable to the Maintenance Plan. It also sets default scheduling parameters.	CHAR	6	M	Maintenance Strategy will be automatically assigned to the Maintenance Item from the Maintenance Plan.
MPOS	WPPOS	Item in the Maintenance Plan	The item in the Maintenance Plan defines the sequence in which the Maintenance Items allocated to the Maintenance Plan are displayed.	NUMC	4	I	Internally populated field

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPOS	PSTXT	Item Short Text	Short description of the contents of the Maintenance Item. Contents of this field are copied to the work order header text	CHAR	40	M	Refer to Data Standard - Short Text
MPOS	TPLNR	Functional Location	Number by which a Functional Location can be clearly identified.	CHAR	23	O	Table Selection.
MPOS	EQUNR	Equipment Number	Number by which a piece of equipment can be clearly identified.	CHAR	18	O	Table Selection.
MPOS	ERNAM	Name of Person who Created the Object	Date on which Maintenance Item was created.	DATS	8	I	Internally populated field
MPOS	ERSDT	Date of creation	Userid of the person who created the Maintenance Item.	CHAR	12	I	Internally populated field
MPOS	AEDAT	Changed On	Blank if Maintenance Item has never been changed or the date on which the Maintenance Plan was last changed.	DATS	8	I	Internally populated field
MPOS	AENAM	Name of Person Who Changed Object	Blank if Maintenance Item has never been changed or the userid of the person to last change the Maintenance Plan.	CHAR	12	I	Internally populated field
MPOS	PLNTY	Task List Type	Describes the associated Task List type: A for General, E for Equipment and T for Functional Location	CHAR	1	M	'A' for General Task List
MPOS	PLNNR	Key for Task List Group	Together with the group counter, the Task List group uniquely identifies a Task List.	CHAR	8	M	Table Selection. The Task List selected must have the same strategy as the Maintenance Plan.

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
MPOS	PLNAL	Group Counter	Part of the key to uniquely identify the Task List.	CHAR	2	M	Table Selection.
MPOS	LTKNZ	Long Text Indicator	Indicates whether long text exists or not.	CHAR	1	I	Internally populated field. 'X' if long text exists, blank if not.
MPOS	WPGRP	Planner Group for Plant Maintenance	Represents the planner group responsible for planning the work order that will be generated from the Maintenance Item.	CHAR	3	M	See Maintenance Item - Planner Group in the attributes section of this document.
MPOS	GEWRK	Object ID of the Work Centre	Internal representation of the main Work Centre.	NUMC	8	M	See Maintenance Item – Main Work Centre in the attributes section of this document.
MPOS	IWERK	Maintenance Planning Plant	A plant in which maintenance tasks are planned and prepared.	CHAR	4	M	See Maintenance Item - Planning Plant in the attributes section of this document.
MPOS	LANGU	Language key of the long text		LANG	1	I	Internally populated field
MPOS	BAUTL	Assembly	A part of a technical object. A technical object can be divided into assemblies that separate it into more clearly defined units.	CHAR	18	O	Table Selection.
MPOS	AUART	Order Type	Type of work order that will be generated when the Maintenance Plan call object is created.	CHAR	4	M	PM02 for Preventive/Predictive Maintenance Orders.
MPOS	ILART	Maintenance activity type	A key for the type of maintenance activity provided.	CHAR	3	M	Table selection

Table	Field	Field Name	Description	Data Type	Len	M/O/C/ I /N	Field Validation Type
			Activities are divided into repairs, shutdowns, regular activities, inspections and so on. In cost evaluations, for example, total costs or number of orders for each technical object, you can classify data according to maintenance activity type.				
MPOS	PSPEL	Work Breakdown Structure Element (WBS Element)	An element of a project.	NUMC	8	N	
MPOS	MITYP	Maintenance Item category	Indicator that determines which maintenance call object the system generates for a due maintenance call.	CHAR	2	M	Only valid when creating a Maintenance Plan manually. Always use 'PM'. This is the default value and will generate a maintenance order.
MPOS	PRIOK	Priority	The key in this field indicates the importance of the processing of the work order.	CHAR	1	M	1 – Emergency 2 – High 3 – Medium 4 – Low 5 – Next Downtime
MPOS	NO_AUFREL KZ	Do Not Release Immediately	Indicator that you can use in the maintenance plan item to control the "Release immediately" status for the order type.	CHAR	1	O	

6. Stakeholder Signoff

Stakeholder Name	Role	Contact	Stakeholder Sign-off
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The Regional Team would like to acknowledge the efforts of the AM Network and Australian SAP PM Business Support Teams in the development of the Master Data Design Standards.

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